
2026 WINTER CLASSIC INVITATIONAL STUDENT CLUSTER COMPETITION ORNL "MYSTERY" CHALLENGE: “QUANTUM TRANSFER LEARNING”

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Logistics

- GitHub Repo:
https://github.com/olcf/hands-on-with-odo/tree/master/challenges/Python_QML_Basics
- Project ID for accessing OLCF Odo and IQM machines:
TRN044
- Timeline:
 - Competition period: March 30 - April 04 2026
 - Submission deadline: 04 April 2026, 5 pm Pacific
 - Mentor support: March 30 - April 02 2026, 9 am - 5 pm Eastern
 - Evaluation/scoring period: 06-10 April 2026
- IQM availability:
 - Two 1-hour slots are reserved on IQM Garnet for the challenge.
 - Thursday April 2nd (3 pm Eastern), Friday April 3rd (3 pm Eastern).
 - Utilize these slots since usage is free during this time and your jobs will wait in a much smaller dedicated queue. During other times, jobs will cost credits and your jobs will be put in a much larger queue and will take longer to start (your script on Odo won't complete in time).
- Need help? Reach out to us in the Slack channel. Note that mentor support will be during 9 am - 5 pm (Eastern) from March 30 - April 02 2026.

Challenge

- Goal: Analyze behavior of a quantum machine learning (QML) code performing transfer learning on simulators and real devices. Make sure to justify your answers by reporting the specific accuracy and runtimes for jobs you ran.
- Objectives:
 1. Parameter sweep
 - Objective: Study the effect that distribution, qubits, and shots have on the accuracy and speed of the application.
 - Complete the following tasks while running on the *lightning.kokkos* device for 30 epochs.
 - Tasks:
 - (a) On one compute node, benchmark the code using 1 through 8 GPUs (MPI tasks) and determine which GPU count is optimal.
 - i. Which configuration leads to the fastest training that still results in $> 90\%$ accuracy? **(2 points)**
 - ii. Which configuration leads to the best accuracy in general? **(1 point)**
 - iii. Which configuration leads to the fastest runtime? **(1 point)**
 - (b) Using 8 GPUs, compare the results when increasing the number of qubits (max 8 qubits).
 - i. Based on the results, do you think using more qubits is beneficial in any way to this application? **(1 point)**
 - (c) Using 3 GPUs and 4 qubits, vary the number of shots.
 - i. How does using shots change things in general? **(1 point)**
 - ii. For a given shot count, monitor the stability of the loss/accuracy across epochs and provide the reason for the observed behavior compared to not using shots. **(2 points)**
 - (d) Using 1 GPU, turn off the quantum network and compare accuracy and speed with the classical version of the code against the quantum method **(2 points)**
 2. Backend evaluation
 - Objective: Load a pre-trained checkpoint of the QML code, and compare the validation phase behavior on a smaller dataset with a real IQM quantum device.
 - Complete the following tasks while using 1 GPU and 5 epochs with the `qml_eval.py` code.
 - Tasks:
 - (a) Comment on the variance of the loss reported across the following methods. What is happening and why?
 - i. *lightning.kokkos* device (shots disabled) **(1 point)**
 - ii. *lightning.kokkos* device (shots=1024) **(1 point)**
 - iii. *qiskit.remote* device (shots=1024), remember to set your IQM token in `keys.sh` **(10 points)**
 3. Expanding the dataset
 - Objective: Expand the dataset to include one more class (ladybugs).
 - Complete the following tasks while using 2 GPUs, 30 epochs, and 4 qubits.
 - Tasks:
 - (a) Create a script called `teamXYZ_expanded.py` by copying and modifying the `qml.py` code to use data from `/gpfs/wolf2/olcf/stf007/world-shared/9b8/hymenoptera_data_expanded`
 - i. What is the "Best Average Loss" reported and which epoch does it occur at? **(2 points)**
 4. Quantum Takeaways
 - Objective: Study the effects that quantum computing has brought to this application.
 - Complete the following tasks based on your experience from past tasks.
 - (a) During optimization, PennyLane executes the circuit multiple times for each image. For 4 qubits it executes things 56 times, while for 5 qubits it executes 70 times. What determines the number of executions (i.e., why 56 times for 4 qubits)? Hint: Think about depth, or try using the `qml.specs` function on the quantum circuit to inspect the number of trainable parameters. **(3 points)**
 - (b) Why does using 4 qubits return 16 different probabilities of outcomes? Why does 5 qubits return 32? **(2 points)**
 - (c) Based on your experience, do you think this is a viable use-case for a QC/HPC application in *this era* of quantum computing? Why or why not? Discuss pros/cons and if you think having a non-cloud quantum computer would be different. There is not necessarily a correct answer here, we just want your team's opinion. **(1 point)**

- Submission instructions
 - Create a submission directory for your team (nominate one person on your team to do this):
`/gpfs/wolf2/olcf/trn044/proj-shared/[teamid]/submission`
 - Make the permission for this directory to user only:
`chmod -R 700 /gpfs/wolf2/olcf/trn044/proj-shared/[teamid]/submission`
 - Place Slurm job output (.out files) into this directory relevant to the answers in your report (see report info below).
 - For "Objective 3", submit your `teamXYZ_expanded.py` Python script in addition to job output.
 - Report: For each task, provide a concise description of the results - these can include, but not limited to, the procedures you followed, any bottlenecks faced, how you overcame them, and answers to the questions posed for the various tasks. Email the report as a single PDF document to `sandovalma@ornl.gov` by submission deadline. **NOTE:** The evaluation will be based on the answers in this report.
- Additional details and tips
 - If you cloned the "Hands on with Odo" github repository prior to March 23rd, you'll need to clone it again or run a (`git pull`) to update things.
 - Do not run on login node. Use interactive (`salloc`) or batch (`sbatch`) jobs.
 - Take advantage of simultaneous job steps to run multiple cases at the same time in a single job script.
 - Only enable/explore the number of shots when explicitly told to do so, otherwise assume no shots.
 - Other than varying the number of GPUs, qubits, and shots, do not modify any other parameters unless explicitly told to do so.
 - Objectives 1-3 can be done in any order – prioritize running on the IQM quantum device from Objective 2 when you can!
 - Reading the original Mari et al. (2019) paper or tutorial may help!
 - The "Quantum Primer" is your friend!

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References

1. Andrea Mari, Thomas R. Bromley, Josh Izaac, Maria Schuld, and Nathan Killoran. Transfer learning in hybrid classical-quantum neural networks. arXiv:1912.08278 (2019).
2. Andrea Mari, "Quantum transfer learning tutorial".
https://pennylane.ai/qml/demos/tutorial_quantum_transfer_learning/
3. Sasank Chilamkurthy, "PyTorch transfer learning tutorial".
https://pytorch.org/tutorials/beginner/transfer_learning_tutorial.html